Simone Romiti

Curriculum Vitae

Personal information

Simone Romiti
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Research activity

My field of research is lattice field theory, a first-principles approach to the non-perturbative regime of quantum field theories. I work in both of its formulations: the Lagrangian one, where observables are computed by Monte Carlo sampling of Euclidean path integrals, and the Hamiltonian one, which is used, among others, by tensor-network and quantum-computing methods. I have made contributions to:

- the solution of the eigenvalue problem of $SU(N)$ lattice Hamiltonians with physics-informed neural networks. In a sole-author paper I introduced an adiabatic training scheme that follows eigenfunctions and eigenvalues along the coupling dependence without an explicit truncation of the Hilbert space [J1]. This is the method that the present proposal develops further.
- the digitisation of gauge fields for quantum simulation. I worked on discrete formulations of the canonical momenta in $SU(2)$ using subsets of the group manifold [J10]. I introduced a discrete transform between the points on the sphere S_3 and their momenta, minimising the breaking of the canonical commutation relations [J9].
- the quantitative matching of the Lagrangian and the Hamiltonian formulations at finite lattice spacing, where they differ by discretisation effects [J4], and its application to the determination of the running coupling of $(2+1)$-dimensional quantum electrodynamics with Monte Carlo and quantum-computing methods [J5].
- the lattice input to the anomalous magnetic moment of the muon within the Extended Twisted Mass Collaboration: the hadronic vacuum polarisation, where I am the reference person of the Bern group in the blinded analysis [C1], [J7], and the hadronic light-by-light contribution, where I am the contact person for the pole contributions and the analysis tool chain [C2].
- the calculation of leading isospin-breaking effects with the RM123 method, in particular the neutron-proton mass difference and the mass splittings of the $\Delta(1232)$ resonances, which was the subject of my doctoral thesis and one of the few theoretical determinations available in the literature [J11], [C10].
- methods and software for lattice calculations: the extraction of multiple exponential signals from Euclidean correlators [J12], the auto-tuning of multigrid solver parameters in the hybrid Monte Carlo [C3], and GPU implementations for gauge-ensemble generation [C9].

Professional experience

01/04/2024 - today	**Postdoctoral researcher**, *University of Bern*, Bern, Switzerland Institute for Theoretical Physics and Albert Einstein Center for Fundamental Physics. Member of the Extended Twisted Mass Collaboration (see Research activity above and Participation in international collaborations below). Supervisor: Prof. Urs Wenger.
01/11/2021 - 31/03/2024	**Postdoctoral researcher**, *University of Bonn*, Bonn, Germany Helmholtz Institute for Radiation and Nuclear Physics (HISKP), Computational Physics group. Member of the Extended Twisted Mass Collaboration (see Research activity above and Participation in international collaborations below). Supervisor: Prof. Carsten Urbach.
01/11/2018 - 31/10/2021	**Doctoral researcher**, *Roma Tre University*, Rome, Italy Department of Mathematics and Physics (see Research activity above). Supervisors: Prof. Vittorio Lubicz, Dr. Silvano Simula.

Education

01/11/2018 - 31/10/2021	**PhD in Theoretical Physics**, *Roma Tre University*, Rome, Italy Thesis: *The neutron–proton mass difference in Lattice QCD+QED*. Supervisors: Vittorio Lubicz, Silvano Simula. Date of submission: 31/10/2021. Date of defence: 22/04/2022. Admitted first in the ranking of the national competitive admission examination.
01/10/2016 - 23/10/2018	**Master's Degree in Theoretical Physics of Elementary Particles**, *Roma Tre University*, Rome, Italy Thesis: *Optimization techniques in the lattice calculation of the hadronic contribution to the muon anomaly*. Supervisor: Prof. Cecilia Tarantino. Final mark: 110/110 cum laude. GPA: 29.85/30. Date of defence: 23/10/2018.
01/10/2013 - 21/07/2016	**Bachelor's Degree in Physics**, *Roma Tre University*, Rome, Italy Thesis: *Astro-Archaeology of the Galactic Centre*. Supervisor: Prof. Giorgio Matt. Final mark: 110/110 cum laude. GPA: 28.84/30. Date of defence: 21/07/2016.

Publications

ORCID: orcid.org/0000-0002-6509-447X **Inspire:** inspirehep.net/authors/1755571
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Peer-reviewed journals

[J1] **S. Romiti**, *$\mathrm{SU}(N)$ lattice gauge theories with physics-informed neural networks*, Phys. Rev. D **113** (2026) 054511, arXiv:2510.26904 [hep-lat].

[J2] T. Jakobs, M. Garofalo, T. Hartung, K. Jansen, P. Ludwig, J. Ostmeyer, **S. Romiti**, C. Urbach, *A comprehensive stress test of truncated Hilbert space bases against Green's function Monte Carlo in* $\mathrm{U}(1)$, Eur. Phys. J. C **86** (2026) 637, arXiv:2510.27611 [hep-lat].

[J3] T. Jakobs, M. Garofalo, T. Hartung, K. Jansen, J. Ostmeyer, **S. Romiti**, C. Urbach, *Dynamics in Hamiltonian lattice gauge theory: approaching the continuum limit with partitionings of* $\mathrm{SU}(2)$, Eur. Phys. J. C **85** (2025) 1418, arXiv:2503.03397 [hep-lat].

[J4] C. F. Groß, **S. Romiti**, L. Funcke, K. Jansen, A. Kan, S. Kühn, C. Urbach, *Matching Lagrangian and Hamiltonian simulations in $(2+1)$-dimensional $\mathrm{U}(1)$ gauge theory*, Eur. Phys. J. C **85** (2025) 1253, arXiv:2503.11480 [hep-lat].

[J5] A. Crippa, **S. Romiti**, L. Funcke, K. Jansen, S. Kühn, P. Stornati, C. Urbach, *Towards determining the $(2+1)$-dimensional Quantum Electrodynamics running coupling with Monte Carlo and quantum computing methods*, Commun. Phys. **8** (2025) 367, arXiv:2404.17545 [hep-lat].

[J6] R. Aliberti *et al.* (Muon $g-2$ Theory Initiative), *The anomalous magnetic moment of the muon in the Standard Model: an update*, Phys. Rept. **1143** (2025) 1–158, arXiv:2505.21476 [hep-ph].

[J7] Extended Twisted Mass Collaboration (C. Alexandrou *et al.*), *Strange and charm quark contributions to the muon anomalous magnetic moment in lattice QCD with twisted-mass fermions*, Phys. Rev. D **111** (2025) 054502, arXiv:2411.08852 [hep-lat].

[J8] Extended Twisted Mass Collaboration (C. Alexandrou *et al.*), *Inclusive hadronic decay rate of the τ lepton from lattice QCD: the $\bar{u}s$ flavour channel and the Cabibbo angle*, Phys. Rev. Lett. **132** (2024) 261901, arXiv:2403.05404 [hep-lat].

[J9] **S. Romiti** and C. Urbach, *Digitizing lattice gauge theories in the magnetic basis: reducing the breaking of the fundamental commutation relations*, Eur. Phys. J. C **84** (2024) 708, arXiv:2311.11928 [hep-lat].

[J10] T. Jakobs, M. Garofalo, T. Hartung, K. Jansen, J. Ostmeyer, D. Rolfes, **S. Romiti**, C. Urbach, *Canonical momenta in digitized $\mathrm{SU}(2)$ lattice gauge theory: definition and free theory*, Eur. Phys. J. C **83** (2023) 669, arXiv:2304.02322 [hep-lat].

[J11] **S. Romiti**, *Leading isospin breaking effects in nucleon and Δ masses*, Rev. Mex. Fis. Suppl. **3** (2022) 0308030, arXiv:2203.04636 [hep-lat].

[J12] **S. Romiti** and S. Simula, *Extraction of multiple exponential signals from lattice correlation functions*, Phys. Rev.

D **100** (2019) 054515, arXiv:1907.09926 [hep-lat].

Conference proceedings

[C1] ETM Collaboration (S. Bacchio *et al.*, incl. **S. Romiti**), *An update on the HVP contribution to $g_\mu - 2$ in isoQCD from ETMC*, 42nd Int. Symp. on Lattice Field Theory (2026), arXiv:2603.03120 [hep-lat].

[C2] N. Kalntis, G. Kanwar, M. Petschlies, **S. Romiti**, U. Wenger, *HLbL contribution to the muon $g-2$ using twisted-mass fermions at the physical point*, 42nd Int. Symp. on Lattice Field Theory (2025), arXiv:2512.21155 [hep-lat].

[C3] M. Garofalo, B. Kostrzewa, **S. Romiti**, A. Sen, *Autotuning multigrid parameters in the HMC on different architectures*, PoS **LATTICE2024** (2025) 276.

[C4] N. Kalntis, G. Kanwar, M. Petschlies, **S. Romiti**, U. Wenger, *Hadronic light-by-light contribution to the muon $g-2$ using twisted-mass fermions*, PoS **LATTICE2024** (2025) 228, arXiv:2412.12320 [hep-lat].

[C5] A. Evangelista *et al.* (Extended Twisted Mass, incl. **S. Romiti**), *Valence leading isospin breaking contributions to $a_\mu^{\text{HVP-LO}}$*, PoS **LATTICE2024** (2025) 260, arXiv:2501.19350 [hep-lat].

[C6] M. Garofalo, T. Hartung, T. Jakobs, K. Jansen, J. Ostmeyer, D. Rolfes, **S. Romiti**, C. Urbach, *Testing the* SU(2) *lattice Hamiltonian built from S_3 partitionings*, PoS **LATTICE2023** (2024) 231, arXiv:2311.15926 [hep-lat].

[C7] M. Garofalo, T. Hartung, K. Jansen, J. Ostmeyer, **S. Romiti**, C. Urbach, *Defining canonical momenta for discretised* SU(2) *gauge fields*, PoS **LATTICE2022** (2023) 040, arXiv:2210.15547 [hep-lat].

[C8] L. Funcke, C. F. Groß, K. Jansen, S. Kühn, **S. Romiti**, C. Urbach, *Hamiltonian limit of lattice QED in* $2+1$ *dimensions*, PoS **LATTICE2022** (2023) 292, arXiv:2212.09627 [hep-lat].

[C9] B. Kostrzewa, S. Bacchio, J. Finkenrath, M. Garofalo, F. Pittler, **S. Romiti**, C. Urbach (ETM), *Twisted mass ensemble generation on GPU machines*, PoS **LATTICE2022** (2023) 340, arXiv:2212.06635 [hep-lat].

[C10] **S. Romiti**, *The neutron–proton mass difference*, PoS **LATTICE2021** (2022) 543, arXiv:2202.01613 [hep-lat].

Published software

[1] **S. Romiti**, *adiabatic_PINNs-suN*: creator. Standalone implementation accompanying the adiabatic-PINN paper. It reproduces the published results.

[2] **S. Romiti** and C. Urbach, *su2_DJT* (2023), Zenodo, doi:10.5281/zenodo.10143897: creator. Archived and citable implementation accompanying the SU(2) digitisation paper.

[3] **S. Romiti**, *lattice_data_tools*: creator and maintainer. Actively maintained toolkit for the production and analysis of lattice data, L-CNN, canonical momenta, solution of Kogut-Susskind Hamiltonian with PINNs, etc.

[4] *su2*: main maintainer. Monte Carlo library for U(1), SU(2) and SU(3) in $d = 2, 3, 4$ dimensions, used for several of the publications above and beyond my own group.

[5] *tmLQCD*: contributor. Community library for twisted-mass lattice QCD, used to generate the ETM collaboration ensembles.

Talks at international conferences, workshops and seminars

30/09-02/10/2026	**Invited speaker.** StaND4LQ 2026, *State of the Art & New Directions for Lattice Gauge Theory and Quantum Simulation*, University of Pisa, Italy. First edition of this workshop, organised entirely by PhD students, bringing together researchers in lattice gauge theories, quantum simulation and quantum many-body physics.
03-07/08/2026	**Invited speaker.** 9th Plenary Workshop of the Muon $g-2$ Theory Initiative, University of Connecticut, Storrs, United States. Talk on the update of $a_\mu^{\text{HVP, isoQCD}}$ from the ETM collaboration.
26/07-01/08/2026	Lattice 2026, the 43rd International Symposium on Lattice Field Theory, University of Maryland, College Park, United States. Talk on SU(N) lattice gauge theories with physics-informed neural networks.
08-13/03/2026	**Invited speaker.** CERN Lattice Seminar, CERN, Geneva, Switzerland. Talk on the adiabatic PINN algorithm for SU(N) lattice gauge theories.
25-27/03/2026	ETM collaboration meeting, Cyprus Institute, Nicosia, Cyprus. Progress of the

	Bern group on the HLbL contribution to $(g-2)_\mu$.
01-05/12/2025	ECT* workshop *Multi-canonical methods and lattice field theory*, Trento, Italy. Talk on the adiabatic learning of the coupling dependence of eigenfunctions and energies.
08-12/09/2025	8th Plenary Workshop of the Muon $g-2$ Theory Initiative, IJCLab, Orsay, France. Talk on behalf of the ETM collaboration on the HLbL contribution.
01-05/09/2025	ECT* workshop *Hamiltonian Lattice Gauge Theories: Status, Novel Developments and Applications*, Trento, Italy. Closing summary talk of the workshop from the organisers, with C. Urbach.
03-07/03/2025	**Invited speaker.** ECT* workshop *Scale Setting: Precision Lattice QCD for Particle and Nuclear Physics*, Trento, Italy. Talk on the sub-permille determination of the lattice spacing in isoQCD within the ETM collaboration.
28/07-03/08/2024	Lattice 2024, the 41st International Symposium on Lattice Field Theory, Liverpool, United Kingdom. Talk on the Nested Sampling algorithm for SU(N) lattice gauge theories (the first of three coordinated talks in that session).
31/07-04/08/2023	Lattice 2023, the 40th International Symposium on Lattice Field Theory, Fermilab, Batavia, United States. Talk on the digitisation of lattice gauge theories in the magnetic basis.
20-24/03/2023	ECT* Gradient Flow Workshop, Trento, Italy. Talk on the gradient-flow determination of the renormalised anisotropy on anisotropic lattices.
08-13/08/2022	Lattice 2022, the 39th International Symposium on Lattice Field Theory, Bonn, Germany. Talk on the Hamiltonian limit of lattice gauge theories.
26-30/07/2021	Lattice 2021, the 38th International Symposium on Lattice Field Theory, MIT, United States (online). Talk on the lattice determination of the neutron–proton mass difference.
26-30/07/2021	Hadron 2021, the 19th International Conference on Hadron Spectroscopy and Structure (online). Talk on the mass splittings of the nucleons and of the $\Delta(1232)$ resonances.

Attendance of international workshops and schools

06-17/07/2026	Lattice@CERN 2026, *Improved estimators for lattice QCD correlation functions*, CERN Theory Department, Geneva, Switzerland.
12-14/02/2025	Zürich Gradient Flow Workshop 2025, University of Zurich, Switzerland.

Awards and grants

2025 - 2026	**Person of Contact** for Computing-time allocation of 22.90×10^6 core-hours on the JUPITER Booster module, awarded by the Gauss Centre for Supercomputing (GCS).
2025 - 2026	**Principal investigator** of a computing-time allocation of 0.20×10^6 core-hours on ALPS at CSCS, the Swiss National Supercomputing Centre, corresponding to 0.80×10^6 GPU node-hours, for the continuum-limit calculation of the η, η' transition form factors on the ETM ensembles.
2026	Winner of a CHF 5,000 grant for the organisation of the *Swiss Lattice Meeting*, the first national gathering for lattice gauge theory in Switzerland, to be held on 14-15/01/2027.
2026	Winner of a CHF 3,000 grant from the *Open Round 2026* programme of the University of Bern, for a visiting research period at TU Wien, Vienna, Austria, from 15/05/2026 to 15/06/2026.
2013 - 2016	Winner of a EUR 3,000 merit scholarship awarded by Roma Tre University for

academic excellence during the Bachelor's degree.

Event organisation

14-15/01/2027	**Organiser** of the *Swiss Lattice Meeting*, one of four organisers, all postdoctoral researchers. First edition of the national meeting for lattice gauge theory in Switzerland, established as a recurring event. I secured the CHF 5,000 of funding for it.
01-05/09/2025	**Leading organiser** of the ECT* workshop *Hamiltonian Lattice Gauge Theories: Status, Novel Developments and Applications*, Trento, Italy. I proposed the workshop and led its organisation over the week.
20-24/01/2025	**Organising support** for the *Workshop on the sign problem in QCD and beyond* (SIGN25), Institute for Theoretical Physics, University of Bern, Switzerland.
2022 - 2024	**Co-organiser**, with M. Mai, of the weekly physics seminars of HISKP, University of Bonn, Germany: invitation of the speakers, chairing of the sessions and broadcasting.
08-13/08/2022	**Organising support** for Lattice 2022, the 39th International Symposium on Lattice Field Theory, Bonn, Germany, as one of the local organisers, including the audio and video broadcasting of the scientific programme.

Participation in international collaborations

01/04/2024 - today	Member of the Extended Twisted Mass (ETM) Collaboration, as reference person of the Bern group for the blinded analysis of the hadronic vacuum polarisation and contact person for the hadronic light-by-light contribution.
01/11/2021 - 31/03/2024	Member of the Extended Twisted Mass (ETM) Collaboration within the Bonn group.
2025 - today	Contributor to the Muon $g-2$ Theory Initiative.

Supervision and teaching

Supervision. Within my postdoctoral positions I have guided graduate students on both the theoretical understanding and the technical execution of their projects, through regular meetings and progress reviews against their milestones.

2024 - today	University of Bern, Switzerland: three PhD students (N. Kalntis, J. F. de la Garza, L. N. Filipovic).
2022 - 2024	University of Bonn, Germany: two MSc and PhD students (C. F. Groß, T. Jakobs, O. Luniachek).

Teaching. Tutorials and assistantships over eight semesters, for six university courses. The teaching material of several of them is maintained in open repositories on GitHub.

01/10/2023 - 01/02/2024	*Computational Physics* (Prof. C. Urbach), University of Bonn, Germany.
01/04/2022 - 31/07/2022	*Quantum Computing* (Prof. L. Funcke), University of Bonn, Germany.
01/10/2021 - 28/02/2022	*Quantum Computing* (Prof. C. Urbach), University of Bonn, Germany.
01/10/2020 - 28/02/2021	*Laboratory of Programming and Calculus* (Prof. S. Bussino), Roma Tre University, Italy.
01/10/2019 - 28/02/2020	*Mathematical Analysis 1* (Prof. U. Bessi), Roma Tre University, Italy.
01/10/2020 - 28/02/2021	*Mathematical Analysis 1* (Prof. L. Chierchia), Roma Tre University, Italy.
01/10/2020 - 28/02/2021	*Fundamentals of Mathematics 2* (Prof. L. Tedeschini Lalli), Department of Architecture, Roma Tre University, Italy, as assistant of the lecturer.
01/02/2020 - 30/06/2020	*Physics 1* (Prof. W. Plastino), Roma Tre University, Italy.

Outreach

15/02/2019	**Co-organiser and presenter** of the public event *Occhi su Marte* ("Eyes on Mars"),

Department of Mathematics and Physics, Roma Tre University, Rome, Italy. The event consisted of seminars, hands-on activities for children and telescope observations for the general public. I contributed to the organisation of the whole event and presented a live demonstration of the Barkhausen effect.

Peer review

2023, 2024	Referee for *Physical Review D*, one manuscript in each year.

Technical skills

Languages	C, C++, Python, Bash, R, SQL, LaTeX, Quarto.
HPC	CUDA, MPI, OpenMP, SLURM, GNU/Linux clusters.
Scientific ML	PyTorch, NumPy, SciPy, SymPy.
Development	Git, GitHub Actions, continuous integration, Docker.
Methods	Monte Carlo and hybrid Monte Carlo, Bayesian inference and model averaging, physics-informed neural networks, variational autoencoders, diffusion models.

Spoken languages

Italian	Mother tongue
English	Fluent
German	Intermediate